AI-Powered Project Evaluator and Viva Examiner for Computing Degree Programs

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Abstract

Currently, It has become very difficult for examiners to conduct project and viva evaluations efficiently with the quick expansion of the number of students. As a result manual evaluations has become a very time consuming process, difficult to scale as well as may lead to inconsistencies due to examiner work load and the human limitations. Therefor this project aims to design and develop an AI assisted system to support and automate the project and viva evaluations to improve the quality of assessments and reduce the examiner burden. The proposed solution for above difficulties is an intelligent evaluation system which allows students to submit project materials , conduct live demonstrations and conduct viva sessions while collecting various types of data such as facial expressions, speech, heart rate measurements to generate accurate feedback of the project and the communication skills of students by using AI techniques such as computer vision, NLP and biometric data analysis. This system provides tools to set rubrics, adjust evaluation settings and review AI generated scores and explanations before finalizing grades so that examiners can reduce their work loads. Therefore Examiners can use this system to manage large number of students more efficiently while ensuring the more consistent and unbiased evaluations. Finally the expected outcome is reliable and scalable AI based evaluation platform that assists examiners in making faster, fairer, and more accurate assessment decision while maintain academic integrity.

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Acronyms

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
BLE	Bluetooth Low Energy
CNN	Convolutional Neural Network
DSR	Design Science Research
FAISS	Facebook AI Similarity Search
HITL	Human-in-the-Loop
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PBL	Project-Based Learning
RAG	Retrieval-Augmented Generation
RNN	Recurrent Neural Network
RST	Rhetorical Structure Theory
SBERT	Sentence-BERT
SHAP	SHapley Additive exPlanations
TrOCR	Transformer-based Optical Character Recognition
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 Background

The rise of Artificial intelligence (AI) and Machine learning (ML) and their fast adoption are changing the higher education sector. As the demand for computer and information technology (IT) professionals continue to surge, universities are experiencing increased enrollment for computing and IT related degree programs. Even though this growth is good, it causes major operational challenges in universities and institutions due to lack of human and technical resources. It has been observed that in most universities, especially in this region, there is a shortage of qualified academic staff in the computer science departments. This has resulted in an imbalance in which a few lecturers are dealing with hundreds of students in multiple modules. It is more challenging for them to maintain high standards of quality and accuracy with such high student-staff ratios. Additionally, the traditional manual evaluation process, particularly for continuous assessments in coding modules and complex final year projects, is time-consuming and labor-intensive [4].

But recent advancements in Natural Language Processing and Large Language Models have made it possible to automate this evaluation process. When analyzing the current progress, we can identify that computers have the potential to evaluate not only objective information but also subjective work such as project reports and codes. It is more than another normal evaluation, and it provides valuable deep information about the performances of the students [9].

The problems identified above are not the only challenges that exist in the context of the application of AI in the educational sector. There is an increasing problem of academic integrity in terms of plagiarism and contract cheating because of the application of AI-generated content. Therefore, there is a need for an efficient evaluation system that can evaluate file based submissions and verifying authorship through an interactive viva test [2].

1.2 Problem Statement

Even though the submission process for universities has been updated, the evaluation process of those submissions is still carried out manually. This study addresses four key problems, with a special emphasis on the realities of university departments in terms of the evaluation process, which are the following,

1. **Staff Gap and Workload:** Evaluating The first problem is the shortage of qualified staff members for the increasing number of assessments. Analyzing the source codes and reports might be time-consuming. Moreover, when only one lecturer must evaluate the assessments of many students within a short period, the process becomes tiring. This will cause lecturers to rush through the evaluation process which affects the quality of the feedback provided to the students [4]. This proposed approach will address this issue by allowing staff to focus only on the initial decision-making process.

2. **The "Black Box" of Evaluation:** Manual grading isn't always transparent. Because the instructor is pressured for time, the student is frequently granted a final grade with no explanation as to reasons for the assignment and deduction of certain grades [5].

3. **Academic Integrity and Verifying Authorship:** With AI tools like ChatGPT, Copilot students can write codes even if they do not understand the logic behind them. So, with available tools there is no way to ensure that the students understand the code they have submitted. Therefore, there is a need for a platform that can "interrogate" the students to ensure authorship [2].

4. **Lack of Personalized Feedback:** In a conventional environment, the viva sessions are time limited. Every candidate is assigned a different level of difficulty by the examiner. This provides little scope for feedback that will help the candidate improve. A standardized and interactive way of AI provides every candidate an equal level of assessment [3].

1.3 Objectives and Scope

1.3.1 Objectives

The goal of this study is to develop a Human-in-the-Loop Decision Support System. The purpose of the above approach is to assist universities, and educational institutions address the shortage of academic staff by implementing a computerized viva and grading procedure that allows lecturers to focus on making final choices. The specific objectives are as follows,

- For Automating the Evaluation of the Reports (Semantic Analysis) : Develop an AI component using Sentence-BERT that will analyze the student reports in a manner as if a human evaluates it. In addition to identifying key words in the report, the new tool will be required to ensure that the "Conclusion" presented by the student is align with the "Research Objectives."

- To Verify Authorship with a Viva exam: Develop an interactive AI viva examiner using Retrieval-Augmented Generation (RAG) which generate technical viva questions and answers by analyzing the students' work and asking those questions from the students. This will enable system to evaluate students' work and verify authorship through the answers given by students.

- Implementation of Multimodal Physiological Monitoring (IoT & Vision): Design a hybrid monitoring system that combines Computer Vision techniques with the use of Internet of Things (IoT) wearable technology. Gaze and Posture: Check gaze and posture of students using a webcam. Physiological Stress: Monitor heart rate variability of a student using smart wearable device like fitness tracker to identify high stress levels during viva examinations which helps the examiner to make decisions.

- To Ensure Transparency (Explainable AI): Normally, in AI when the data is provided to get a result it will analyze that and give a result. But it does not know how it gets to that final score because it thinks silently inside a box. So for escape this 'black box' grading issue, there is a need for a usaging of the Explainable Scoring System. Unlike a conventional AI model that simply provides the result as a number with no context, this system provides an explanation for the grade given. It gives a detailed explanation that traces the grade back to the corresponding reasons in the form of pointing out an error in the code or the omission in the report. It allows the students to know exactly why they received the grade they did and instills confidence in the system [5].

1.3.2 Scope

This project has the following boundaries,

- Domain: The system architecture of our application is modular. It can be used for any module that requires submission of codes related files and documents.

- Technical Implementation: The system will be built as a web application, with Next.js for the frontend and Django for the backend and will use open-source AI models (Hugging Face/Whisper).

- Multimodal Analysis Scope: To improve the quality of the viva exam and the exam environment the system will use multimodal monitoring procedure which analyzes the visual behavior and physiological information of the student.
  Visual Behavioral Analysis – A web cam is used for analyzing the visual behaviors of the student such as eye tracking and head posing. From that visual behavior information, the system will identify student's engagement throughout the session and possible misbehaviors like cheating.

Stress Monitoring – The system will use a fitness tracker worn on the student's wrist to get the real time biometric data of the student such as heart rate. Then system will analyze Heart Rate Variability during the exam and from that data examiner can detect any suspicious activity during the session.

- Evaluation Mode: The system only acts as Human-in-the-Loop Decision Support System. It does not decide final grades independently instead it suggests grades according to provided metrics which help examiners to determine final grades. So, the final judgement will be made by the examiner. Also, the system will generate an overall report about the full session which includes recommended scores by the system, student's emotional details, explainable ai part which justify the final grade and any suspicious acts.

Chapter 2

Literature Review

2.1 Previous Work

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2.1.1 Viva Evaluation

The evaluation of oral responses in higher education has evolved significantly with the integration of artificial intelligence.

Recently, several studies have been conducted to facilitate physically disabled people in writing examinations using voice-based answer evaluation systems. These systems take automatic speech recognition (ASR) as input, capturing the students' voice responses before converting them into text that undergoes deep analysis. The resultant text is analyzed using some semantic and syntactic analysis techniques for relevance, correctness, and completeness according to some marking scheme or rule. More importantly, T5, LDA (Latent Dirichlet Allocation), and RST(Rhetorical Structure Theory) enable a higher comprehension of the content since they mark the product fairly and consistently under minimal human intervention.[1].

Over the past years LLMs (Large Language Models) have been used in enhancing viva-based assessment by enabling context-aware and interactive questioning. Specifically, the LLM-based system TAMIGO, proposed by an Indian university proved highly efficient and effective in generating viva questions with . This study shows that this system has helped reducing question repetition and information leakage across viva sessions.[2].

Generally, the ability to assess beyond what a student is saying during the presentation is common in the case of AI systems. MOSAIC-F is a multi-modal system that is able to interpret cues based on heart rate, eye movement, body language, etc., in order to assess the communication skills and confidence level of students[3]. These combined with peer/presenter feedback can provide a personal feedback for non-verbal communication assessment. It portrays the importance of using multi-modal analysis in the case of Viva assessment.

2.1.2 Code and Technical Document Evaluation

When it comes to assessment and code evaluation, checking only for the structure and wording is not enough. There should be a method or technique to check whether the actual meaning is correct.

A study done by A. Ayaan and K.W. Ng proposes a hybrid grading system that addresses the above issue. Conventional NLP similarity measures like Jaccard smilarity, Cosine similarity and Edit distance are used for the comparison of surface level similarity. Then it uses deep semantic embeddings to understand the meaning of the text. These semantic embeddings are developed by models such as TensorFlow's Universal Sentence Encoder. This assists in making appropriate assessments regarding the evaluations submitted by students if they have been described in different words but with a similar meaning[4].

There are also several studies on the use of TrOCR (Transformer-based Optical Character Recognition) for digitizing handwritten answer papers. A study by Balkrishna Rasiklal demonstrates an effective solution for large-scale exam grading with minimized examiner bias using this model. First, the content is converted into machine-readable text. Then the GPT-based models evaluates the content against a predefined marking scheme (methodologies), facilitating scalable assessment[5].

Not like OCR, TrOCR uses a Transformer encoder-decoder architecture that departs from conventional CNN-RNN models. It uses a pre-trained vision encoder and language decoder. And the speciality is that the same TrOCR architecture can be used for all printed, handwritten and scene text and is good at handwritten text where OCR struggles[6].

Also, the use of LLMs has been increased over the past years in automating code and assesment evaluations. These LLMs can create code summaries and detect if there are any logical flwas , and provide feedback with reasoning. [2].

2.1.3 Automated Grading

Many automated pipelines have been incorporated in rule-based scoring layers in order to avoid unfair decisions in the grading of assesments. When the system is not confident about a student's answer, it does not make a final decision on its own. Instead it serves as a guide for a teacher or examiner to review manually.

Although there is a huge advancement in AI-based assessment systems lately, studies on this area show that many of these systems focus mainly on technology rather than education. These models are designed in a way to maximize the accuracy, but they do not consider how students learn and what the learning objectives are. Therefore, a gap might be created between the educational values and the technical accuracy.[7]. Also, according to research conducted by the University of Washington and Hensun Innovation, it shows that human interaction is essential when deciding the final grades. Therefore it cannot be totally replaced the human evaluation with AI systems [8].

2.1.4 Bias Detection and Coursework Authentication

Not only in the automation of the evaluation process, but aslo in ensuring the fairness and integrity of the assessments, AI can be used. Based on the research conducted, techniques like Genetic Algorithms and explainable AI methods like

SHAP (SHapley Additive exPlanations) to study grading patterns from different instructors. By analysing these things, the hidden biases and inconsistencies are detected and identifying these issues helps in improving grading rubics and subjectivity.[9].

AI systems can also be used in plagiarism detection. According to a study done by Mausumi Sahu on Plagiarism Detection Using Artificial Intelligence Technique in Multiple Files , it shows that KNN (K-Nearest Neighbor) machine learning algorithm can be used to compare text across multiple files. [10].

And authentication of the coursework can be done using tools like AutoViva. Here, the depth and clarity of a student's submission is compared with their spoken answers during an AI generated Viva session. If the mismatch is very high, the system flags it as a possible sign of cheating. [11].

2.2 Gaps in Literature

2.2.1 Fragmented Assessment Systems

Despite significant advances in AI-driven student assessment, existing systems provide isolated services such as automated text grading, voice-based answer evaluation, or standalone code evaluation. There is a significant absence of a unified platform that integrates all these services into a single cohesive framework. This fragmentation limits the ability to evaluate both student output and overall understanding.

2.2.2 Weak Human-in-the Loop Design

Most of the existing evaluation systems do not adequately balance automation process with the human judgement in academic assesments. This may cause problems with the accuracy of the grades. Therefore, there is a significance need for a AI grading tool where the examiner decides the final grade.

2.2.3 Absence of Real-Time Adaptive Viva Interaction

Existing viva-based assessment tools primarily rely on predefined or randomly generated questions. These systems do not support real-time adaptive questioning that responds dynamically to a student's spoken answers. As a result, they fail to emulate the interactive probing and clarification processes used by human examiners during technical viva examinations.

This work proposes a comprehensive and human-in-the-loop evaluation framework using AI for speech semantic analysis tasks like viva interactions, hallucination feedback generation, pedagogically aligned evaluation, and bias aware evaluation of computer degree programs as a solution to bridge those gaps.

Chapter 3

Methodology

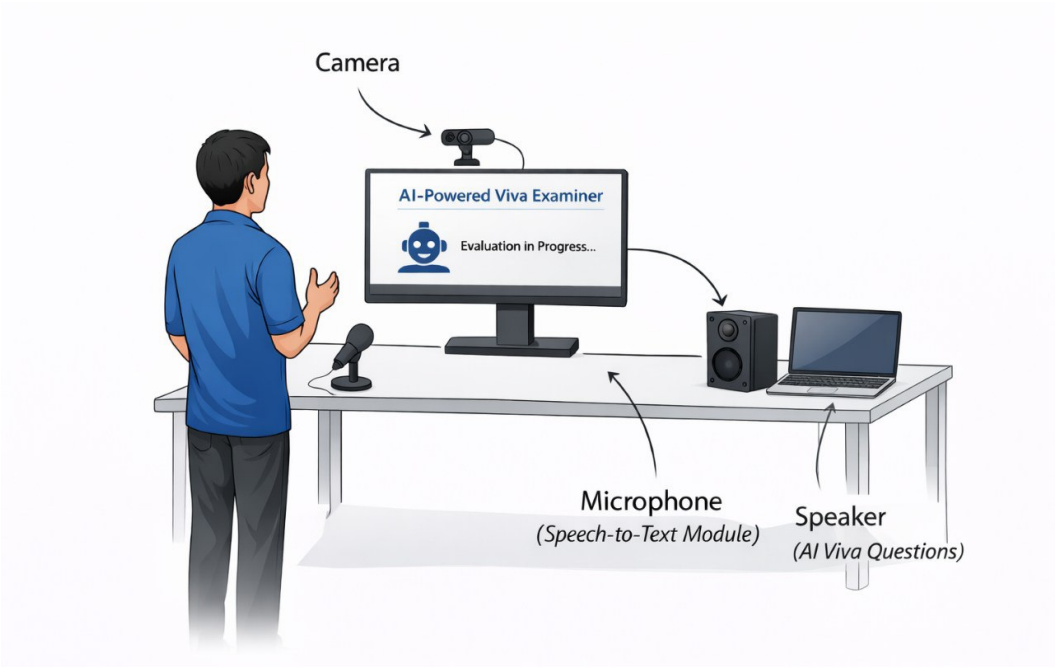

Figure 3.1: Physical setup.

This chapter presents the methodological framework used for the development, implementation, and validation of the proposed AI-powered Project Evaluator and Viva Examiner. The methodology is comprises of two main sections each describing the system architecture, analytical models and the multi-model data acquisition strategy used to support academic evaluation while retaining the human oversight over the system.

3.1 Research Design

This proposed research adopts a Design Science Research (DSR) methodology to design, develop, and evaluate an AI-supported, human-in-the-loop academic evaluation system intended for general academic modules. The system is not restricted to a specific discipline or academic level; instead, it is designed as a general-purpose evaluation framework applicable to project-based assessments, oral examinations,

and practical evaluations within academic institutions. Unlike the traditional oral viva, the proposed system mimics the role of a physical evaluator with AI-powered integrations to facilitate an environment where the student can be evaluated during the viva. The setup includes a camera, a microphone, and a speaker, along with a BLE band to facilitate the communication and detection between the system proposed and the student [11]. All of the interactions are recorded, and a report is generated to provide subtle details of the assessment to the evaluator in order to facilitate and maintain a transparent and reviewable environment where the evaluator has the final decision making [7].

The system integrates computer vision, natural language processing (NLP), static program analysis, BLE sensors to backup subtle details, AI powered question generation that are aligned with the pre-determined rubrics and metrices provided by the evaluator, and a grading system to provide an overall understanding about the performance of the student [3]. Although AI creates and structures the questions and performs evaluations, the evaluator always has the final authority of the grading of the student making this assessment outcome being complaint with any academic, institutional regulations, ethical considerations and quality standards.

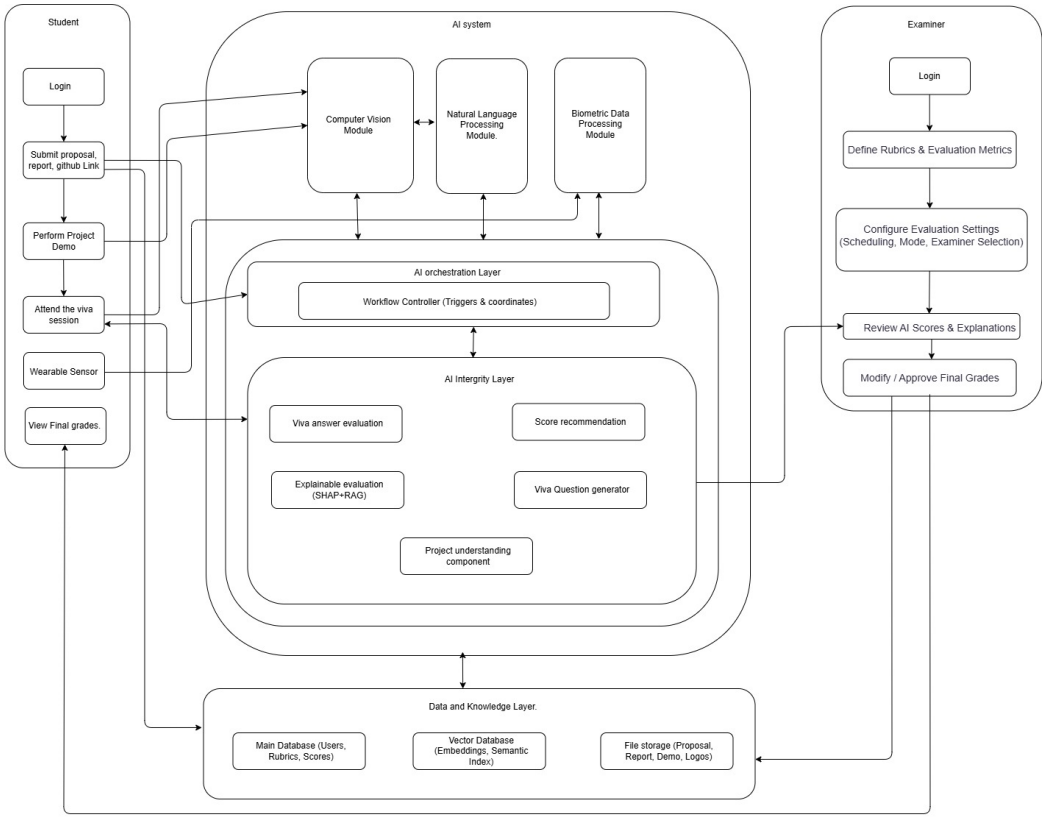

Figure 3.2: System Architecture

As shown in the above figure 3.2, the overall system architecture is modular, meaning, independent evaluation of project reports, source code, presentation and the viva performance, can be either together or separately evaluated. Outputs from each module are aggregated using a weighted score fusion mechanism where the evaluator can configure the weights based on the rubrics of the

modul[4].Thereby, this mechanism promotes modularity, transparency and flexibility of the evaluation grading and explicitly eradicates black-box grading.

Sentence-BERT (SBERT) is employed to generate the semantic embedding of textual project reports [4]. Semantic similarity techniques are used to detect plagiarism of reports and check its originality. A Cosine similarity is proposed to be implemented to determine the gap between the original rubric descriptions with the report of the student on various criteria such as the problem formulation, methodology, and technical depth and interpretation.

SonarQube is used to analyze the quality of the code and statically analyze the alignment with the metrics. Metrics including complexity, modularity, cohesion, commenting, duplication are extracted from the code of the student and normalized to the rubrics defined by the evaluators in order to maintain consistent quality in assessing across different programming languages.

Retrieval-Augmented Generation (RAG) framework is used in designing the viva examination module. The syllabus content defined by the rubrics and metrics by the evaluator and any datasets related to the project are indexed to a FAISS vector database. This relevant contextual information is retrieved in real time during the viva session and fed to the proposed large language model GPT-5.2 to generate well structured and aligned questions guided by Bloom's Taxonomy [2]. Thereby, the system is able to generate adaptive questions based on the answers of the student maintaining the flow and depth of the viva conversation, eventually evolving dynamically.

The viva examination component is designed using a Retrieval-Augmented Generation (RAG) framework. Project-specific datasets, syllabus content, and instructor-defined grading rubrics are embedded using SBERT and indexed in an FAISS vector database. During the viva session, relevant contextual information is retrieved in real time and provided to the large language model GPT-5.2 to generate structured, rubric-aligned questions guided by Bloom's Taxonomy [2]. This adaptive questioning mechanism enables the depth and direction of questioning to evolve dynamically based on the student's explanations and demonstrated understanding.

Whisper model of OpenAI is used to transcribe the response from the students during the viva examination. Semantic relevance is also taken into consideration when determining the relevance of the responses of the student with the rubrics configured by the evaluator. For this specifically, the project proposes a transformer-based cross-encoder framework. Apart from the technical level evaluation, the system uses speech-level features including the delay in responding, the frequency of pauses between responses and the clarity as measurement of confidence and preparation of the student. However, these said no-technical evaluation is explicitly excluded from the primary grading system and is only provided to the evaluator as a supplementary to the performance of the student [3].

With the above proposed contextual, visual, and speech-based analyzes, the project proposes to integrate physiological indicators for emotion and behavior assessment in detail. A Bluetooth Low Energy (BLE) heart rate monitor band is used during the viva session. This is proposed to be worn by the student during the evaluation and is time-synchronized to the audio and video recording pipeline. To improve the accuracy of the results, a short period of this pre-measured and recorded while on the student's hand as a baseline period. Thereafter, the rela-

tive deviations during technical questioning are analyzed as auxiliary indicators of cognitive load and stress. These are presented to the examiner as contextual decision supporting information. Overtime, this information is guaranteed to present important decision making in defining rubrics and context to the module.

In addition to visual and speech-based analyses, physiological indicators are incorporated to support emotion and behavior assessment. Student heart rate is continuously monitored using a Bluetooth Low Energy (BLE) heart-rate band during the viva session, with measurements time-synchronized to the audio–video recording pipeline. A short baseline period is recorded at the beginning of the session, after which relative heart-rate variability, peak deviations during technical questioning, and recovery duration are analyzed as auxiliary indicators of cognitive load and stress. These physiological signals are not included in the weighted score fusion process but are presented to the examiner as contextual decision-support information.

Explainable Artificial Intelligence (XAI) mechanisms are integrated in the system to enhance reasoning, transparency and trust. Evidence backed feedback generated by LLM GPT-5.2, constrained by pre-determined rubrics is used to reason the evaluation scoring of the student's performance, closing linking the actual performance with the assessment critieria. For code based quality scoring, feature important analysis is used while attention based explanations are used to highlight any specific sections of the project report that are vital in the final assessment.

Previous studies show that although XAI is vital in backing automated system and significantly increases the transparency and the ability to interpret, the human oversight is essential throughout the whole system [7]. Thereby, the evaluator has the authority over the system to validate AI-generated questions, recommendations, inspect the reasoning, and rewatch the recorded evidences and finally adjust and approve the final grading. All of these examiner interactions are also proposed to be logged in order to maintain transparency and accountability of the system.

3.2 Data Collection

This project combines information from the software code, the written documentation, and human interaction through a multimodal approach. This is beneficial as it allows the project to assess all aspects of the students' work, from the output to their ability to explain how it works.

Student Artifacts for Technical Evaluation

For assessing students' technical skills, three categories of learning artifacts are collected by this system. Project reports submitted in PDF or DOCX format are used for assessing students' understanding of problems and their conclusions. Source code submitted by students can be obtained from their Git repositories. This helps in assessing students' code quality and implementation logic. Rubrics provided by lecturers are used as learning artifacts and are considered to check if students are assessed for learning outcomes.

Behavioral and Interaction Data (Viva)

To make the viva exam as close to an actual project defense as possible, interaction data is captured in the oral viva exam. Audio input is used to capture the responses to be processed in speech-to-text engines and analyzed semantically to assess the quality of the responses' reasoning, and video input is captured and analyzed to detect non-verbal cues like eye gaze and head pose, as proposed in the MOSAIC-F framework [3]. This is helpful in engagement analysis and possible issues of academic integrity.

Data Storage and Ethical Compliance

Once the data is collected, it is stored in a manner that allows it to be quickly accessed and analyzed properly. The text data as well as the source code is passed through a FAISS vector database. Student metadata is stored in a PostgreSQL database. This allows semantic search within the RAG pipeline.

Chapter 4

Timeline and Resource Required

4.1 Timeline

Table 4.1: Project Timeline and Development Phases

Project Phase	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**	**M7**	**M8**
Literature Review & Requirement Analysis	✓	✓						
System Architecture & Design	✓	✓						
Database Design & Setup		✓	✓					
Project Understanding Module (NLP)			✓	✓				
Viva Question Generation (RAG)				✓	✓			
AI-Assisted Viva Evaluation					✓	✓		
Computer Vision Integration					✓	✓		
Explainable AI & Score Recommendation						✓	✓	
System Integration						✓	✓	
Testing, Validation & Iteration							✓	✓
Evaluation, Analysis & Documentation							✓	✓

4.2 Resources Required

This system requires the below mentioned hardware resources for the development phase of the system. To maintain accuracy and effieciency of the system,

certain quality standard levels of the devices as show below are expected. The specifications and costs are summarized below.

4.2.1 Hardware Resources

- **Web Camera**:Ugreen 35636 camera is used to capture video during the viva sessions. This camera is able to capture footage at 2K 30fps quality and is available at an affordable price in the market. High quality 2K video capturing will provide accurate facial landmark detection.
- **Microphone**:Razer Seiren Mini Condenser microphone is used to capture high-quality conversions. This is an on-desk microphone which can be used even for group evaluations and is capable of capturing subtle noises.
- **Speaker**:HAVIT SK208 PRO speaker is used to deliver the generated questions to the students during the oral examinations. This is affordably available in the market.
- **Heart Rate Sensor**: Xiaomi Smart Band 9 is one of the BLE devices that is having a high accuracy to cost ratio. This device provides real time hear rate variation information to provide auxiliary information to the examiner.
- **Webcam Tripod**: A basic webcam tripod that is capable of holding the webcam at a configurable angle and height in a stable position is sufficient.

4.2.2 Estimated Budget

Table 4.2 presents the estimated budget for the required hardware resources.

Table 4.2: Estimated Hardware Resource Budget

Resource	**Estimated Cost (LKR)**
Ugreen 35626 Web Camera	13,950
Razer Seiren Mini Microphone	21,500
Xiaomi Smart Band 9	8,950
HAVIT SK208 PRO Speaker	5,000
Webcam Tripod	1,500
Total Estimated Cost	**50,900**

The total cost of LKR 50,900 as shown above in the above table 4.2 shows that this development phase can be implemented affordabaly and easily with accessible equipment without going for high cost devices.

Chapter 5

Conclusion

This System aims on solving a huge practical and important problem that faced by examiners when evaluating a large number of students. Due to the computer engineering expansion every year, it will be very time consuming, stressful to the human examiner to conduct project evaluations and viva sessions manually. In addition to that the workload for the lecturer is increased as well as this sometimes leads to inconsistencies in grading due to fatigue or limited time. These will be caused to delayed in result processing.Therefore, there is a strong need for a more efficient, more reliable and more smarter method that can support examiners and improve the overall academic assessment process. Therefore by introducing an AI assisted evaluation platform, this system provides a modern and innovative solution to these challenges. The system supports examiners by project analyzing, conducting viva sessions, and recommending unbiased score in a transparent, well organized and systematic way. Therefore it is clear that, this system helping for reducing human error, minimizing bias, and ensuring that students are evaluated more fairly based according to the predefined rubrics and objective criteria. Furthermore, examiners are remaining in control of the final grading decisions, which maintains trust and accountability in the academic process. In addition, the system saves a huge amount of time for lecturers. Furthermore , allowing lecturers to focus more on teaching, mentoring, and curriculum development rather than spending excessive time on manual grading tasks. Overall, this project demonstrates how AI can be meaningfully applied to enhance the academic evaluation process more efficiently. Additionally this system can be improved to support more subjects, different evaluation styles as well as larger number of students with further development, testing, and real-world implementation. In the future, Many educational institutions has the potential to be adopted, helping to modernize the education system by making student assessment faster, more reliable, and easier to manage while maintaining academic integrity, fairness, and quality in education.

References

[1] A. Pimpalshende, S. N. Peddineni, V. Srirama, D. Thalla, and I. Katukuri, "Voice based answer evaluation system for physically disabled using natural language processing," *Journal of Information Systems Engineering and Management*, vol. 10, no. 30s, p. –, 2025.

[2] Anishka, D. Sethi, N. Gupta, S. Sharma, S. Jain, U. Singhal, and D. Kumar, "Tamigo: Empowering teaching assistants using llm-assisted viva and code assessment in an advanced computing class," 2024. arXiv preprint.

[3] A. Becerra, D. Andres, P. Villegas, R. Daza, and R. Cobos, "Mosaic-f: A framework for enhancing students' oral presentation skills through personalized feedback," 2025. Preprint.

[4] A. Ayaan and K.-W. Ng, "Automated grading using natural language processing and semantic analysis," *MethodsX*, vol. 14, p. 103395, 2025.

[5] B. R. Yadav, "Ai-driven exam evaluation systems: Challenges, innovations, and future directions," *International Journal of Electronics Automation*, vol. 02, no. 02, pp. 7–13, 2024. Available online.

[6] M. Li, T. Lv, J. Chen, L. Cui, Y. Lu, D. Florencio, C. Zhang, Z. Li, and F. Wei, "Trocr: Transformer-based optical character recognition with pre-trained models," 2021.

[7] V. González-Calatayud, P. Prendes-Espinosa, and R. Roig-Vila, "Artificial intelligence for student assessment: A systematic review," *Applied Sciences*, vol. 11, no. 12, p. 5467, 2021.

[8] Z. V. Tian, A. Liu, L. Esbenshade, S. Sarkar, Z. Zhang, K. He, and M. Sun, "Implementation considerations for automated ai grading of student work," 2025.

[9] J. Divasón, F. J. Martínez De Pisón Ascacibar, A. Romero, and E. Sáenz-de Cabezón, "Artificial intelligence models for assessing the evaluation process of complex student projects," *IEEE Transactions on Learning Technologies*, vol. 16, no. 5, pp. 694–707, 2023.

[10] M. Sahu, "Plagiarism detection using artificial intelligence technique in multiple files," *International Journal of Scientific & Technology Research*, vol. 5, no. 04, pp. —, 2016.

[11] H. Cao and Collaborator, "Automated viva voce using generative ai for student coursework authentication," in *Proceedings of the Conference (ACM)*, 2025.